

Preparation of a nickel complex $[\text{Ni}(\text{NH}_3)_6]\text{Cl}_2$ and analysis of metal complexes using UV-VIS spectrophotometry

In this experiment, preparation and analysis of a simple Ni(II) complex, $[\text{Ni}(\text{NH}_3)_6]^{2+}$ will be carried out. The nature as well as strength of the ligand is one of the major factors that determine the formation and stability of a metal complex. From a comparative study of the degree of splitting of the d-orbitals by various ligands for a metal species in a particular oxidation state, one can arrange the well-known and widely used ligands in the order of increasing strength. This arrangement of ligands in the increasing order of strength is called the *spectrochemical series*.



A stronger ligand can replace a weaker ligand from a metal center and therefore ammonia can easily replace OH^- or H_2O from a metal center. Similarly, ethylenediamine (en) can replace ammonia and in this case the additional stability due to chelation also drives the reaction forward. The changes in enthalpy and entropy on formation of ethylenediamine (en) complexes starting with $[\text{Ni}(\text{NH}_3)_6]^{2+}$ indicates that contribution from entropy is the deciding factor for chelate effect. In the reactions of nickel chloride hexahydrate with ammonia, it has been known that initially there can be formation of nickel hydroxide (observed as a green precipitate), which reacts further with ammonia to give ammine complexes. It has also been shown by separate experiments that nickel hydroxide reacts with ammonia to give ammine complexes of Ni(II). It may also be noted that the reaction is dependent upon the stoichiometry and if excess of ammonia is not used, one can end up getting a mixture of products with varying compositions e.g. $[\text{Ni}(\text{NH}_3)_5(\text{H}_2\text{O})]\text{Cl}_2$, $[\text{Ni}(\text{NH}_3)_3(\text{H}_2\text{O})_3]\text{Cl}_2$, $[\text{Ni}(\text{NH}_3)_5\text{Cl}]\text{Cl}$ etc. To make sure that the right complex has been made, one has to estimate the amount of nickel, ammonia and chloride in the complex. (Ammonia can be estimated by acid base titration and chloride can be estimated gravimetrically by converting it to AgCl with AgNO_3 . These are however, not taken up in this experiment). Nickel can be estimated by three well known methods: (a) by spectrophotometric analysis (wherein the ammine complex is converted back to the aqua complex and its absorbance compared with standard solutions of nickel at a specific wavelength of $\sim 395 \text{ nm}$) (b) by gravimetric analysis after converting nickel to its bis(dimethylglyoximate) complex and (c) by complexometric titration with EDTA.

Preparation of the $[\text{Ni}(\text{NH}_3)_6]^{2+}$ complex

The complex is prepared by taking nickel chloride hexahydrate as a water solution to which an

excess of aqueous ammonia is added. The color of the solution changes from pale dark green to purple. The solution is then cooled in an ice bath upon which the $[\text{Ni}(\text{NH}_3)_6]^{2+}$ complex will be precipitated out as purple crystals. This can be separated by filtration, washed with ammonia solution and dried.

PROCEDURE

(A) Preparation of Hexamminenickel(II) chloride (*to be done in groups of two*)

1. Take 5 ml of a solution of Nickel chloride hexahydrate (containing about 6 g of nickel chloride) in a 100ml beaker.
2. Take 10 ml of conc. aqueous ammonia in another beaker.
3. Add the aqueous ammonia slowly to a rapidly stirred solution of the nickel chloride.
4. Make sure that the color of the solution has changed from pale green to intense purple (if not, add more of ammonia).
5. Allow the solution to stand at room temperature for 1-2 minutes, cover it with filter paper or aluminum foil and then cool it in a crushed ice bath without disturbance for about 15 minutes.
6. If purple-blue crystals of $[\text{Ni}(\text{NH}_3)_6]\text{Cl}_2$ has not formed try initiating crystallization by gently scratching the sides of the beaker with a glass rod.
7. Decant carefully the solution or filter it to separate the crystals
8. Dry the crystals by keeping it between a few pieces of filter paper.
9. Determine the weight of the dried complex and calculate the percentage yield.

Most transition metal complexes are colored. This implies that as white light is passed through a sample of the complex, some of the wavelengths of the visible spectrum are absorbed by the sample and the rest transmitted, so the complementary color is imparted to the complex (see below).

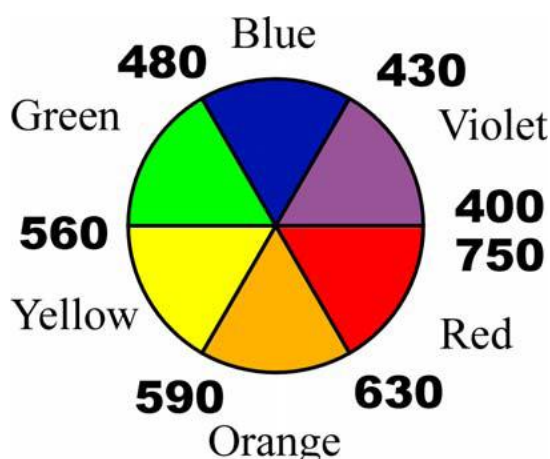


Figure 1: An artist's wheel

TABLE 1: COLOR ABSORBED AND COLOR TRANSMITTED (complementary color)

COLOR OF LIGHT ABSORBED	APPROX. λ RANGES/nm	COLOR OF LIGHT TRANSMITTED
RED	700-620	GREEN
ORANGE	620-580	BLUE
YELLOW	580-560	VIOLET
GREEN	560-490	RED
BLUE	490-430	ORANGE
VIOLET	430-380	YELLOW

The Crystal Field Theory (CFT) replaced Valence Bond Theory (VBT) in a major way to explain the observed colours (spectral properties) and magnetic properties of various metal complexes. Some salient features of CFT have been provided below:

- Only the electrostatic interactions between the central metal ion and the ligands are considered and no orbital overlap is supposed to take place.
- Initially, the 5 d-orbitals are equivalent in energy but on approach of ligands the energy increases for all the 5 orbitals as a result of electrostatic repulsion between the electron cloud of the central metal ion and the ligands, treated as point charges. The *barycentre* thus moves higher.
- When the ligands approach nearer and position themselves as to give the complex a definite geometry for e.g. octahedral, tetrahedral, square planar etc. the degeneracy of the d-orbitals is lifted.
- In octahedral complexes 2 sets are formed ---- a lower energy t_{2g} set comprising d_{xy} , d_{yz} and d_{zx} (these point in between the ligands) and a higher energy e_g set comprising $d_{x^2-y^2}$ and d_{z^2} (these point directly at the ligands and therefore face higher repulsion.)
- In case of tetrahedral arrangement the splitting order is just there reverse. The e set is lower in energy and the t_2 set is higher. Here the 'g' has been dropped because tetrahedral complexes are *not centrosymmetric*.
- The energy difference between the sets is termed as Δ_o or $10Dq$ in the language of CFT. The energies involved in the d-d transitions are actually a measure of this crystal field splitting. The magnitude of this splitting depends on several factors, the most important being the nature of the metal (size, oxidation state) and the nature of the ligand. The geometry is also important; lesser the number of ligands lesser is the crystal field splitting. The value of Δ_o is almost half in tetrahedral than in octahedral.

Variation in Δ_o values with varying ligands for the same central metal ion gives rise to different colours (*absorption spectra*) of metal complexes; based on this variation, a series called the spectrochemical series has been developed. It arranges the ligands in the increasing order of the crystal field splitting exerted by them.

The UV-Vis absorption spectrum of nickel complex ions

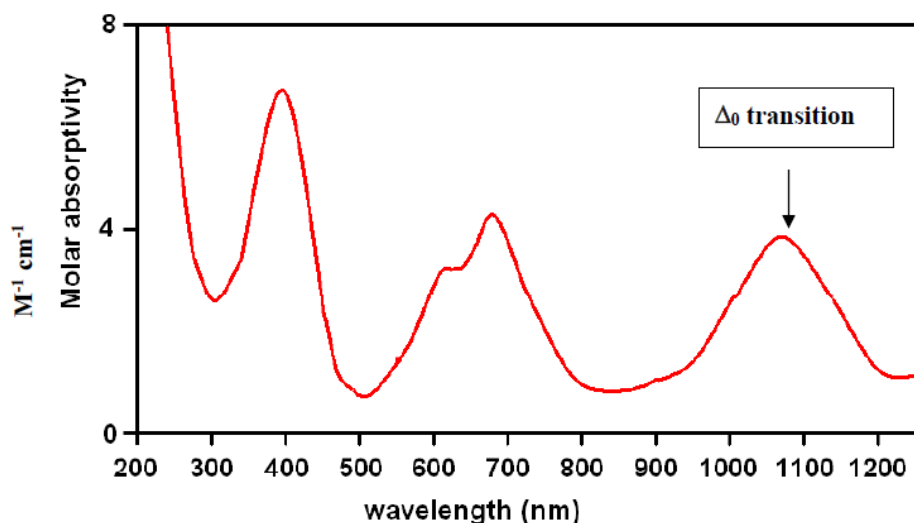


Figure 2: Absorption Spectrum of [Ni(H₂O)₆]²⁺ (note that the absorption band corresponding to the Δ_0 value of this complex occurs beyond the range of the spectrometer, that is, above 1000 nm)

Explanation of colour of [Ni(NH₃)₆]²⁺:

The hexaamminenickel(II) complex ion, [Ni(NH₃)₆]²⁺ appears pale blue due to strong absorption in the yellow region, which is complementary to blue. It shows two main absorption peaks, with maxima at ~360 nm and ~590 nm. The first peak extends over the near-ultraviolet and violet regions of the visible spectrum.

Recording the UV-Vis spectra of [Ni(H₂O)₆]²⁺ and [Ni(NH₃)₆]²⁺

Prepare 0.1(M) solution of [Ni(H₂O)₆]²⁺ and [Ni(NH₃)₆]²⁺ using 10 mL volumetric flask. Record the spectra in a UV-Vis spectrophotometer of each of the solutions between wavelengths 340-1000 nm after proper mixing. For your information, molecular weight of [Ni(H₂O)₆]Cl₂ and [Ni(NH₃)₆]Cl₂ are 237.69 g/mol and 231.779 g/mol respectively.

Questions and Discussion

1. Write the equation for the synthesis reaction of [Ni(NH₃)₆]²⁺, report the yield, and calculate the percent yield.
2. Report the λ_{max} values for [Ni(H₂O)₆]²⁺ and [Ni(NH₃)₆]²⁺
3. Describe the observed colors of each complex and explain these observations in terms of their λ_{max} values.
4. Provide a step-by-step procedure for all experiments conducted today.